A deep learning cell classification and sorting platform for single cell high-dimensional morphology characterization

Kevin B Jacobs[1], Andreja Jovic[1], Kiran Saini[1], Anastasia Mavropoulos[1], Ryan Chow[1], Simo Zhang[1], Esther Lee[1], Michael Phelan[1], Jeanette Mei[1], Janifer Cruz[1], Chassidy Johnson[1], Nianzhen Li[1], Thomas J. Musci[1], Mahyar Salek[1], Maddison (Mahdokht) Masaeli[1]

[1]Deepcell Inc. Menlo Park, CA Contact: kevin@deepcellbio.com

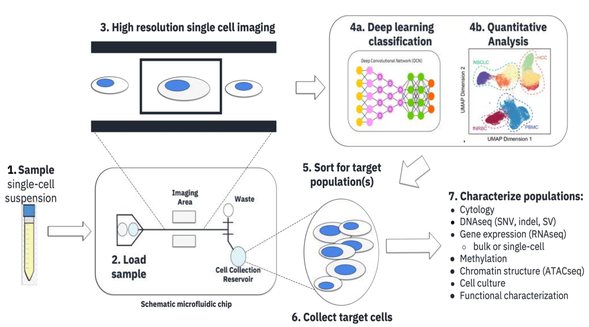
→ Introduction

Although cell morphology is often the clinical gold standard for diagnosis and prognosis for many diseases and conditions, cell morphology information relies on low throughput and semi-quantitative approaches, and has seen limited application in combination with comprehensive molecular and functional characterization methods. This dissonance is due to the largely manual and subjective process of collecting morphology information and limitations of cell sorting methods which can damage or perturb cells.

We present a platform to address these limitations based on a novel microfluidic optical device capable of high-throughput cell imaging and sorting. The hardware is complemented by a machine learning infrastructure capable of real time analysis on cell images to generate high-dimensional morphologic descriptors and classifications, a machine learning assisted human image annotation tool to facilitate model training, a comprehensive atlas of pre-labeled cell images, and a library of pre-trained models for specific biological applications. Our platform yields populations of cells that are label-free, viable, and minimally perturbed, allowing sorted cells to be recovered and further characterized by molecular and functional assays. Additionally, cell images can be used to generate high-dimensional morphological profiles to reveal and explore previously unrecognized heterogeneous cell populations.

Our platform is able to sort cells to enrich for these morphology-based subpopulations, which can be further characterized to link high-dimensional morphology to molecular profiles for deeper understanding of disease complexity.

→ Deepcell platform schematic



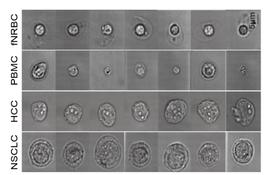
(1) A single cell suspension of sample is **(2)** loaded onto the microfluidic chip and **(3)** as they flow past the imaging area, high resolution bright-field images are captured and the images are used by the **(4a)** AI-classifier for deep learning cell classification and/or **(4b)** quantitative morphological analysis using the high-dimensional single cell morphological data. The target cells identified by the AI-classifier (4a) are **(5)** sorted by routing them to the Cell Collection Reservoir. The enriched cells can be **(6)** retrieved and **(7)** processed for downstream analysis including cytological, functional or molecular characterization.

→ Machine learning infrastructure

The Deepcell platform utilizes machine learning models to classify and generate high-dimensional quantitative morphological descriptions of cells. Our model architecture is based on the InceptionV3 (Szegedy et al. 2016) convolutional neural network architecture, modified for grayscale images and to output high-dimensional morphological descriptors (often called "embeddings" in the machine learning literature).

We have implemented this architecture to run in real-time on our instrument which allows images to be fit by previously trained models and generates classification and high-dimensional morphology descriptions. If cell sorting is desired, the model outputs are used to to determine whether to discard or retain each cell and, if retained, which collection well to route each cell.

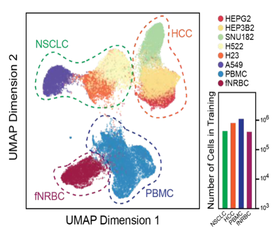
→ Visualizing quantitative morphology



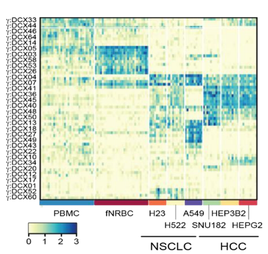
Shown to the left are representative images of non-small cell lung cancer (NSCLC), hepatocellular carcinoma (HCC), peripheral blood mononuclear cells (PBMC) and fetal nucleated red blood cells (fNRBC) generated by the Deepcell platform and used for model training in the example below.

Cell images are assigned cell class labels and high-dimensional morphology descriptors (embeddings). These vectors are not generally interpretable in terms of conventional morphology metrics, but can be used to perform cluster analysis to group morphologically similar cells and visualized using tools like Uniform Manifold Approximation and Projection (UMAP, McInnes et al. 2020) and clustered heatmaps.

This UMAP plot depicts high-dimensional morphology vectors projected onto a two dimensional space. Each point represents a single cell and the color indicates the true cell type. This projection show distinct clustering of NSCLC cell lines (H522, H23, A549), HCC cell lines (HEPG2, HEP3B2, SNU182), healthy donor PBMCs, and fNRBCs.



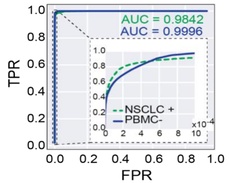
This heatmap is another representation of high-dimensional cell morphology vectors where each column represents a single cell in the respective cell classes and each row is an embedding vector. Each cell class exhibits distinct coordinate combinations that can be used to understand morphological properties unique to cell classes.



→ Malignant cell identification & enrichment

To demonstrate the platform's capabilities, we imaged cells and trained a neural network classifier to identify and enrich malignant cells from NSCLC dissociated tumor cell samples. We verified classifier identification of malignant cells by comparing the percent malignant cells identified by the classifier to EpCAM expression. Enrichment of malignant cells identified by the classifier was confirmed by RNA and DNA analysis.

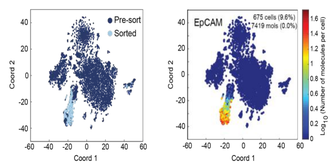
The performance of the NSCLC tumor cell classifier was evaluated *in silico*. Receiver operating characteristic curves for classification of NSCLC tumor cells in a PBMC background. The area under the curve (AUC) for positive selection is 0.9842 and negative selection is 0.9996.

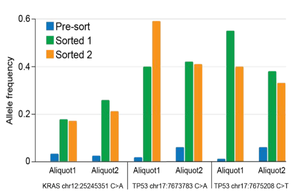


Sample	Fraction of Malignant Cells		
	EpCAM (+)		Deepcell Classifier (+)
	Protein	RNAseq	Morphology
1	1.3%	3.3%	2.2%
2	11%	11.6%	12%
3	69%	39%	40%

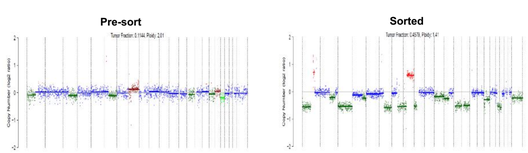
NSCLC disassociated tumor samples with varying fraction of malignant cells, as determined by EpCAM cell-surface protein markers and RNA expression, were used to evaluate performance of classifier to identify malignant cells. Morphology- based enrichment yielded comparable fraction of malignant cells.

Single cell RNA sequencing shows high levels of EpCAM expression in the sorted cell population (in light blue) confirming epithelial cells enrichment.





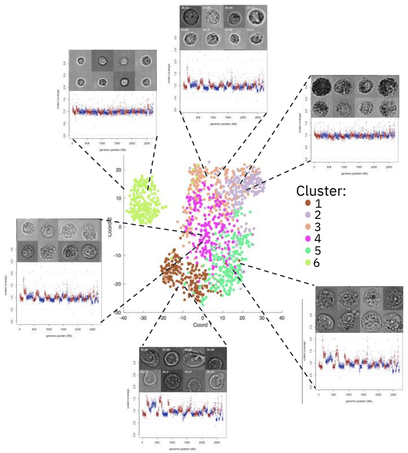
DNA point mutations in *KRAS* and *TP53* genes were detected and had increased allele frequency in sorted compared to pre- sorted samples that was consistent between instruments and replicates.



Whole-genome DNA Copy number variation (CNV) analysis was performed on a pre- and post-sorted NSCLC disassociated tumor sample. Increased sensitivity of CNVs is detected in sorted samples confirming enrichment of malignant cells based on morphology alone.

→ Exploring morphological heterogeneity

High-dimensional quantitative morphological analysis of the NSCLC malignant cells identified by the classifier was performed to assess heterogeneity. UMAP analysis demonstrates the presence of 6 clusters of malignant cell subpopulations. CNV analysis on the subpopulations demonstrates distinct morphology and CNV profiles between the clusters.



High-dimensional quantitative morphological analysis of NSCLC malignant cells (clusters 1-6) by UMAP. Each point represents a cell and is colored based on cluster assignment. The model was trained to sort for the respective clusters and CNV analysis was performed on the subpopulations. Morphological heterogeneity and subtle differences in CNV profiles between the malignant cell subpopulations were observed.

→ Conclusions

- The Deepcell platform enables high-throughput cell classification, sorting and high-dimensional morphological analysis based on morphology delivering unperturbed and intact cells compatible with further downstream analysis.
- Presented is a use case where the neural network classifier was trained to identify and enrich for malignant cells from NSCLC dissociated tumor cell samples.
  - Morphology-based classification resulted in comparable estimates of malignant cell fractions compared to alternative protein and gene expression approaches.
  - Tumor cell enrichment was demonstrated by increased allele frequency and sensitivity of CNV in sorted compared to pre-sorted samples.
  - Morphological and molecular tumor cell heterogeneity was observed and studies are ongoing to further explore this link.
- The ability to perform high-dimensional morphological analysis, enrich based on morphological features, and perform molecular analysis on the morphologically distinct subpopulations demonstrates the power of the Deepcell platform to explore tumor heterogeneity and discover new insights into tumor biology.